

DATA STRUCTURES

LIST

```
In [4]: l=[]  
1
```

```
Out[4]: []
```

```
In [7]: l=list()  
1
```

```
Out[7]: []
```

```
In [10]: #List is ordered seq of items
```

```
In [12]: l1=[1,2,3,4]  
l1
```

```
Out[12]: [1, 2, 3, 4]
```

```
In [15]: li=[1,0.2,1+3j,True,'new'] #Heterogenous  
li
```

```
Out[15]: [1, 0.2, (1+3j), True, 'new']
```

```
In [18]: #Length  
my_list=[10,20.5,"Hello"]  
len(my_list)
```

```
Out[18]: 3
```

```
In [20]: #Nested List
```

```
In [22]: nest=[1,[3,4],3]  
print(nest)
```

```
[1, [3, 4], 3]
```

```
In [24]: len(nest)
```

```
Out[24]: 3
```

```
In [26]:  #List Indexing
```

```
In [28]:  nest[0]
```

```
Out[28]: 1
```

```
In [30]:  nest[1]
```

```
Out[30]: [3, 4]
```

```
In [32]:  nest[1][0]
```

```
Out[32]: 3
```

```
In [34]:  nest[2][0]
```

```
-----  
---  
TypeError                                Traceback (most recent call last)  
~\AppData\Local\Temp\ipykernel_20552\2770975312.py in <module>  
----> 1 nest[2][0]
```

```
TypeError: 'int' object is not subscriptable
```

```
In [37]:  a=[1,3,[4,5,[8,"hello",6]],45,99]  
a  
a[3]
```

```
Out[37]: 45
```

```
In [39]:  #List are Mutable
```

```
In [41]:  #Mutable : value of elements of a list can be altered in same object
```

```
In [43]:  a=[1,2,3]  
print(a)  
  
id(a)
```

```
[1, 2, 3]
```

```
Out[43]: 2637984926848
```

```
In [45]: ▶ a[1]=20
          print(a)

          id(a)
```

```
[1, 20, 3]
```

```
Out[45]: 2637984926848
```

```
In [47]: ▶ #List slicing
```

```
In [49]: ▶ numbers=[10,20,30,40,50,60,70,80]
          numbers[0:4]
```

```
Out[49]: [10, 20, 30, 40]
```

```
In [52]: ▶ #List Concatination
          l1=["dsad",'b','c']
          l2=['a',6,4.0]
          l3=l2+l1
          print(l3)
```

```
['a', 6, 4.0, 'dsad', 'b', 'c']
```

```
In [54]: ▶ #List Methods
```

```
In [56]: ▶ #List.append()
```

```
In [58]: ▶ my_list=[10,20,30,40]
          my_list.append(50)
          my_list
```

```
Out[58]: [10, 20, 30, 40, 50]
```

```
In [60]: ▶ lst=[1,2,3,4,5,6]
          lst.append([8,9])
          lst
```

```
Out[60]: [1, 2, 3, 4, 5, 6, [8, 9]]
```

```
In [63]: ▶ #List.extend() -it adds individual item at end
```

```
In [65]: ▶ lst=[1,2,3,4,5,6]
          lst.extend([8,9])
          lst
```

```
Out[65]: [1, 2, 3, 4, 5, 6, 8, 9]
```

```
In [67]: l=[1,2,3,4,5,6]
l.extend([1,[4,9],24,[9,90]])
l
```

```
Out[67]: [1, 2, 3, 4, 5, 6, 1, [4, 9], 24, [9, 90]]
```

```
In [73]: #List.insert()
lst=['one','two','four']
lst.insert(2,"three")
print(lst)
```

```
['one', 'two', 'three', 'four']
```

```
In [89]: #List.remove()-Remove based on value
numbers=[10,20,30,40,10]
numbers.remove(10)
numbers
```

```
Out[89]: [20, 30, 40, 10]
```

```
In [78]: #List.pop()-Remove based on index position
numbers=[10,20,30,40]
numbers.pop(0)
numbers
```

```
Out[78]: [20, 30, 40]
```

```
In [79]: numbers=[10,20,30,40]
del numbers[0:3]
numbers
```

```
Out[79]: [40]
```

```
In [81]: #List.clear()
a=[1,2,3,4]
a.clear()
print(a)
```

```
[]
```

```
In [83]: a=[1,2,3,4]
a.clear()
a.extend([5,6,7,8])
print(a)
```

```
[5, 6, 7, 8]
```

```
In [86]: > #List.reverse()
```

```
In [90]: > l=[1,4,9,[4,6]]
l.reverse()
l
```

```
Out[90]: [[4, 6], 9, 4, 1]
```

```
In [92]: > #List.sort()
```

```
In [95]: > lst=[1,20,5,5,4.2]
lst.sort()
lst
```

```
Out[95]: [1, 4.2, 5, 5, 20]
```

```
In [97]: > lst=[1,20,5,5,4.2]
lst.sort(reverse=True)
lst
```

```
Out[97]: [20, 5, 5, 4.2, 1]
```

```
In [99]: > l=['aef','b','a','f']
l.sort()
l
```

```
Out[99]: ['a', 'aef', 'b', 'f']
```

```
In [101]: > lst=[1,20,'b',5,'a']
print(lst.sort())
```

```
-----
---
TypeError                                Traceback (most recent call la
st)
~\AppData\Local\Temp\ipykernel_20552\4275344380.py in <module>
      1 lst=[1,20,'b',5,'a']
----> 2 print(lst.sort())

TypeError: '<' not supported between instances of 'str' and 'int'
```

```
In [105]: > lst=[1,20,5,4.2]
sorted(lst)
```

```
Out[105]: [1, 4.2, 5, 20]
```

In [107]: `l1`

Out[107]: `[1, 20, 5, 4.2]`

In [111]: `#l1.count()`

In [113]: `numbers=[1,2,3,1,1,0,3,4,2,5]`
`print(numbers.count(1))`

3

In [115]: `#Shallow Copy / Deep Copy`

In [117]: `#Shallow copy`

In [135]: `l1=[1,2,3]`
`l2=l1`
`l2.append(4)`
`print(l1)`
`print(l2)`

`[1, 2, 3, 4]`
`[1, 2, 3, 4]`

In [129]: `id(l1)`

Out[129]: 2637984932352

In [130]: `id(l2)`

Out[130]: 2637985800576

In [123]: `#Deep Copy/Carbon copy`

In [127]: `l1=[1,2,3]`
`l2=l1.copy()`
`l2.append(4)`
`print(l1)`
`print(l2)`

`[1, 2, 3]`
`[1, 2, 3, 4]`

In [131]: `id(l1)`

Out[131]: 2637984932352

```
In [133]: ▶ id(12)
```

```
Out[133]: 2637985800576
```

TUPLES

```
In [138]: ▶ tup1=()  
tup1
```

```
Out[138]: ()
```

```
In [140]: ▶ type(tup1)
```

```
Out[140]: tuple
```

```
In [142]: ▶ tup2=tuple()  
tup2
```

```
Out[142]: ()
```

```
In [145]: ▶ t=(1,0.2,1+3j,True,'new',[1,4]) #Heterogenous  
t
```

```
Out[145]: (1, 0.2, (1+3j), True, 'new', [1, 4])
```

```
In [148]: ▶ #Nested Tuple  
tup=('siva',25,25.0,1+2j,True,[50,100],[99,22,33])  
tup
```

```
Out[148]: ('siva', 25, 25.0, (1+2j), True, [50, 100], (99, 22, 33))
```

```
In [150]: ▶ #Tuple Indexing
```

```
In [152]: ▶ tup[0]
```

```
Out[152]: 'siva'
```

```
In [154]: ▶ tup[-1][2]
```

```
Out[154]: 33
```

```
In [156]: ▶ #Tuple Slicing
```

```
In [158]:  tup[0:3]
```

```
Out[158]: ('siva', 25, 25.0)
```

```
In [160]:  #Tuples are Immutable
```

```
In [166]:  mytuple=(1,2,3,4,5)
```

```
In [167]:  mytuple[0]
```

```
Out[167]: 1
```

```
In [169]:  del mytuple[0]
```

```
-----  
---  
TypeError                                Traceback (most recent call la  
st)  
~\AppData\Local\Temp\ipykernel_20552\1004790074.py in <module>  
----> 1 del mytuple[0]  
  
TypeError: 'tuple' object doesn't support item deletion
```

```
In [171]:  mytuple[0]=10
```

```
-----  
---  
TypeError                                Traceback (most recent call la  
st)  
~\AppData\Local\Temp\ipykernel_20552\24554413.py in <module>  
----> 1 mytuple[0]=10  
  
TypeError: 'tuple' object does not support item assignment
```

```
In [174]:  del mytuple
```

```
-----  
---  
NameError                                Traceback (most recent call la  
st)  
~\AppData\Local\Temp\ipykernel_20552\72011595.py in <module>  
----> 1 del mytuple  
  
NameError: name 'mytuple' is not defined
```

```
In [176]: #Interview Question
```

```
In [178]: tup=(1,2,3,[4,5],(10,11,12))
```

```
In [182]: tup[3][0]=40
```

```
In [184]: tup
```

```
Out[184]: (1, 2, 3, [40, 5], (10, 11, 12))
```

```
In [186]: tup[3]=50  
tup
```

```
-----  
---  
TypeError                                Traceback (most recent call la  
st)  
~\AppData\Local\Temp\ipykernel_20552\2099372669.py in <module>  
----> 1 tup[3]=50  
      2 tup  
  
TypeError: 'tuple' object does not support item assignment
```

```
In [188]: #Tuple Methods
```

```
In [189]: mytuple1=('one','two','three','four','one','one','two','three')
```

```
In [194]: mytuple1.count("one")
```

```
Out[194]: 3
```

```
In [195]: mytuple1.index('one')
```

```
Out[195]: 0
```

```
In [198]: mytuple=(43,67,99,12,6,90,67)  
tuple(sorted(mytuple))
```

```
Out[198]: (6, 12, 43, 67, 67, 90, 99)
```

```
In [200]: mytuple
```

```
Out[200]: (43, 67, 99, 12, 6, 90, 67)
```

SETS

```
In [202]: ➤ set_var=set()  
          print(set_var)  
          type(set_var)
```

```
set()
```

```
Out[202]: set
```

```
In [204]: ➤ a={10,30,20,40,5,'a'}  
          print(a)
```

```
{'a', 20, 5, 40, 10, 30}
```

```
In [206]: ➤ s={10,4,20,20,30,30,30,5}  
          print(s)
```

```
{4, 5, 20, 10, 30}
```

```
In [208]: ➤ print(s[1])
```

```
-----  
---  
TypeError                                Traceback (most recent call la  
st)  
~\AppData\Local\Temp\ipykernel_20552\819424605.py in <module>  
----> 1 print(s[1])
```

```
TypeError: 'set' object is not subscriptable
```

```
In [212]: ➤ #Creating set by List
```

```
In [214]: ➤ l=[1,2,3,1,2,3,1,2]  
          s=set(l)  
          print(s)
```

```
{1, 2, 3}
```

```
In [216]: ➤ #Creating set by Tuple
```

```
In [218]: ➤ t=('a','b','c')  
          tup_set=set(t)  
          tup_set
```

```
Out[218]: {'a', 'b', 'c'}
```

In [220]: `#Set Methods`

In [222]: `c=set()
c`

Out[222]: `set()`

In [224]: `#set.add()`

In [242]: `c.add(3)
c`

Out[242]: `{3, 4, 5}`

In [228]: `#set.updat()
s={3,4}
s.update([5,6,8])
print(s)`

`{3, 4, 5, 6, 8}`

In [231]: `#set.copy()
c={3,4,5}
d=c.copy()
d.add(6)
print(d)
print(c)`

`{3, 4, 5, 6}`
`{3, 4, 5}`

In [233]: `#set.discard()`

In [235]: `s={1,2,3,5,4}
print(s)`

`{1, 2, 3, 4, 5}`

In [240]: `s.discard(4)
print(s)`

`{1, 2, 3, 5}`

In [243]: `s.discard(7)
print(s)`

`{1, 2, 3, 5}`

In [245]: `#set.remove()`

In [247]: `s.remove(10)`

```
-----  
---  
KeyError                                Traceback (most recent call la  
st)  
~\AppData\Local\Temp\ipykernel_20552\1259242377.py in <module>  
----> 1 s.remove(10)  
  
KeyError: 10
```

In [250]: `s={1,5,2,3,6}
s.clear()
print(s)`

`set()`

In [252]: `s={1,5,2,3,6}
del s
print(s)`

```
-----  
---  
NameError                                Traceback (most recent call la  
st)  
~\AppData\Local\Temp\ipykernel_20552\734359141.py in <module>  
      1 s={1,5,2,3,6}  
      2 del s  
----> 3 print(s)  
  
NameError: name 's' is not defined
```

Set Operations

In [255]: `set1={1,2,3,4,5}
set2={3,4,5,6,7}`

In [259]: `#Union`

In [260]: `set1.union(set2)`

Out[260]: {1, 2, 3, 4, 5, 6, 7}

In [262]: `#Intersection`

In [264]: `set1.intersection(set2)`

Out[264]: {3, 4, 5}

In [266]: `set1-set2`

Out[266]: {1, 2}

In [268]: `set2-set1`

Out[268]: {6, 7}

In [271]: `set1.symmetric_difference(set2)`

Out[271]: {1, 2, 6, 7}

In [273]: `c={3,4,5}`
`d={3,4,5,6}`

In [275]: `c.issubset(d)`

Out[275]: True

In [277]: `d.issubset(c)`

Out[277]: False

In [278]: `d.issuperset(c)`

Out[278]: True

Dictionary

In [281]: `d={}`
`print(d)`
`type(d)`

{}

Out[281]: dict

```
In [283]: >>> d=dict()  
          print(d)  
          type(d)
```

```
{}
```

Out[283]: dict

```
In [285]: >>> dict={1:456,10.4:[3,9,15], 'key3':(94,8,23), 'key4':{'item1','item2','item2'  
dict
```

Out[285]: {1: 456,
10.4: [3, 9, 15],
'key3': (94, 8, 23),
'key4': {'item1', 'item2', 'item4'}}

```
In [287]: >>> #Nested Dictionary
```

```
In [290]: >>> car_type={'car1':1989, 'car2':123, 'car3':{'1':3}}  
          print(car_type)
```

```
{'car1': 1989, 'car2': 123, 'car3': {1: 3}}
```

```
In [292]: >>> car_type["car2"][1]
```

```
-----  
---  
TypeError                                Traceback (most recent call la  
st)  
~\AppData\Local\Temp\ipykernel_20552\2685029258.py in <module>  
----> 1 car_type["car2"][1]  
  
TypeError: 'int' object is not subscriptable
```

```
In [294]: >>> car_type["car3"][1]
```

Out[294]: 3

```
In [296]: >>> marks={'history':45, 'Geogrpby':54, 'Hindi':56}  
          marks
```

Out[296]: {'history': 45, 'Geogrpby': 54, 'Hindi': 56}

```
In [298]: >>> len(marks)
```

Out[298]: 3

```
In [300]: marks.keys()
```

```
Out[300]: dict_keys(['history', 'Geogrphy', 'Hindi'])
```

```
In [302]: marks.values()
```

```
Out[302]: dict_values([45, 54, 56])
```

```
In [304]: #Indexing
```

```
In [306]: marks['history']
```

```
Out[306]: 45
```

```
In [308]: #Dictionaries are mutable
```

```
In [310]: marks['english']=47 #Add single value  
marks
```

```
Out[310]: {'history': 45, 'Geogrphy': 54, 'Hindi': 56, 'english': 47}
```

```
In [312]: marks.update({'chemistry':89,'Physics':98})  
marks #Add multiple values
```

```
Out[312]: {'history': 45,  
          'Geogrphy': 54,  
          'Hindi': 56,  
          'english': 47,  
          'chemistry': 89,  
          'Physics': 98}
```

```
In [314]: #RePlace value
```

```
In [327]: marks['Hindi']=64  
marks
```

```
Out[327]: {'history': 45,  
          'Geogrphy': 54,  
          'english': 47,  
          'chemistry': 89,  
          'Physics': 98,  
          'Maths': 64,  
          'Hindi': 64}
```

```
In [329]: #RePlace Key
```

```
In [330]: marks['Maths']=marks.pop('Hindi')
marks
```

```
Out[330]: {'history': 45,
           'Geogrphy': 54,
           'english': 47,
           'chemistry': 89,
           'Physics': 98,
           'Maths': 64}
```

```
In [332]: #delete item in dictionary
```

```
In [333]: del marks['english']
```

```
In [335]: marks
```

```
Out[335]: {'history': 45, 'Geogrphy': 54, 'chemistry': 89, 'Physics': 98, 'Maths':
64}
```

RANGE

```
In [338]: range(10)
```

```
Out[338]: range(0, 10)
```

```
In [340]: list(range(10))
```

```
Out[340]: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
```

```
In [342]: tuple(range(10))
```

```
Out[342]: (0, 1, 2, 3, 4, 5, 6, 7, 8, 9)
```

```
In [344]: list(range(8,11))
```

```
Out[344]: [8, 9, 10]
```

```
In [346]: list(range(0,11,2))
```

```
Out[346]: [0, 2, 4, 6, 8, 10]
```

```
In [347]: list(range(8,0,-1))
```

```
Out[347]: [8, 7, 6, 5, 4, 3, 2, 1]
```

